Materials Today: Proceedings 42 (2021) 2680–2685

Contents lists available at ScienceDirect

Materials Today: Proceedings

journal homepage: www.elsevier.com/locate/matpr

Polyphosphates as thermal stabilizers for poly(vinyl chloride)

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ARTICLE INFO

Article history:
Available online 10 February 2021

Keywords:
Polyvinyl chloride
Polyphosphates
Thermal stability
Weight loss
Thin films

ABSTRACT

Polyphosphates had been used as thermal stabilizers for polyvinylchloride (PVC). The thermal stability of PVC films stabilized with polyphosphates was verified by using weight loss method, Fourier transform infrared (FTIR) spectroscopy, thermal aging test, optical microscope and atomic force microscope. The outcomes demonstrated that mixing of polyphosphates with PVC can obviously enhance the stability time of PVC and improve the initial color of it. Furthermore, polyphosphates could considerably decrease the concentrations of forming double bonds within the PVC chemical structure and decrease weight loss. In addition, polyphosphate has an excellent ability to reduce the releasing of hydrochloric acid (HCl) molecules. Hence polyphosphates can replace the labile chlorine atoms within the chemical structure of PVC chains.

1. Introduction

One of the most important synthesized polymers in the world is polyvinyl chloride (PVC) due to its amazing diverse features, easy to synthesis, low cost, flexibility and light weight [1,2]. Thus, the polymer chemical structure undergoes elimination reaction with releasing HCl molecules and formation double bond when processing at high temperature. This is because of the thermal degradation which affects the chemical and physical properties of the PVC such as change the color and strength of the polymer [3,4]. Hence the PVC color is changing quickly from light color to darker colors such as dark brown or black [5]. Many approaches were used to address this problem and the researchers exhibited that the best way is mixing the PVC with thermal-stabilizers to stop the thermal-degradation. This considers a very good practical method and easy to apply. Yet various thermal stabilizers have been used in both academic and industry areas to reduce the degradation and the most popular stabilizers are organometallic materials, lead salts, organic polymers, and metal soaps [6]. The main job for the stabilizer is to react with the elaborated hydrochloric molecules or to interact with PVC backbone to stop alkene group formation [7].

The most famous stabilizer is the metal soap thermal stabilizer such as zinc soap and calcium soap stabilizers. Hence the Administration of Food and Drug in United State has recommended these types of stabilizers as nontoxic and safe materials to use as thermal stabilizer with PVC [8]. From this point of view, it is important to design new materials work as thermal stabilizers and in the same time they are not toxic or safe to utilize. In recent times, researchers have studies polyols as thermal stabilizers such as pentaerythritol [4,9]. However, this type of stabilizer sometimes causes of poor color and formation of "plate-out". Even though many authors have exhibited metal alkoxide due to they have a brilliant ability with capturing hydrochloride molecules such as zinc-glycerol alkoxide [10,11]. In most applications that require thermal treatment of the plastic wastes, such as waste-to- energy or waste-to-chemical, hydrochloric acid (HCl) is released from chlorinated compound (mostly polyvinyl chloride - PVC) in the waste materials [21]. For these energy and chemical applications, hydrochloric acid must be removed, or reduced drastically as it is corrosive [22], can degrade catalysts [23]. As the source of chlorine is mostly from

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https://doi.org/10.1016/j.matpr.2020.12.704
2214-7853/